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Project Synopsis

Title: AI-Based Multi-Platform Product Analysis and Recommendation System

INTRODUCTION:

In today's digital world, online shopping has become very common. Many e-commerce platforms such as Amazon, Flipkart, Myntra, and Meesho offer a wide range of products with different prices, discounts, ratings, and customer reviews. However, users often find it difficult to compare products across multiple platforms and choose the best one.

This project aims to develop an AI-Based Multi-Platform Product Analysis and Recommendation System that collects product data from different platforms, analyses it using data analytics, and provides smart recommendations using artificial intelligence. The system helps users make better purchasing decisions by comparing multiple factors such as price, rating, discount, and customer feedback in one place.

PROBLEM STATEMENT:

Online shoppers face several challenges while purchasing products:

- They need to visit multiple websites to compare the same product.
- Prices and discounts vary across platforms.
- Large numbers of customer reviews are difficult to understand.
- Users get confused in choosing the best product.
- There is no single system that combines price comparison, review analysis, and intelligent recommendation.

OBJECTIVES:

- To collect product data from multiple e-commerce platforms.
- To compare products based on price, rating, discount, and reviews.
- To clean and preprocess the collected data.
- To perform sentiment analysis on customer reviews using AI.
- To calculate a product score based on multiple factors.
- To recommend the best product to the user.
- To provide a simple and user-friendly interface.

METHODOLOGY:

The project follows a step-by-step process:

1. User Input

The process begins with the user interacting with the system through a simple and user-friendly interface.

- The user enters the product name (e.g., *mobile phone, shoes, laptop*).
- The system may also allow optional filters such as price range, brand, or category.
- This input acts as the primary query for searching products across different platforms.

2. Data Collection

Once the input is received, the system collects product-related data from multiple e-commerce platforms such as Amazon, Flipkart, and others.

- Data is gathered using **APIs** (if available) or **web scraping techniques**.
- The collected data includes:
 - Product name
 - Price
 - Discount
 - Ratings
 - Customer reviews
 - Product links
- This step ensures that real-time and relevant product data is available for analysis.

3. Data Preprocessing

The raw collected data is often unstructured and inconsistent. Therefore, preprocessing is performed to improve data quality.

- Removal of duplicate product entries
- Handling missing or incomplete values
- Converting data into a uniform format (e.g., price in same currency)
- Cleaning unwanted symbols or text
- Structuring data into tables (using tools like Pandas)

This step ensures that the dataset is clean, reliable, and ready for analysis.

4. Feature Extraction

After preprocessing, important attributes (features) are extracted from the dataset.

- Key features include:
 - Price
 - Rating
 - Discount percentage
 - Number of reviews
- Additional derived features may be calculated, such as:
 - Price after discount
 - Popularity score (based on reviews count)

These features play a crucial role in evaluating and comparing products.

5. Sentiment Analysis

Customer reviews are analyzed using AI techniques from Natural Language Processing.

- Reviews are classified into:
 - Positive
 - Negative
 - Neutral
- Techniques used:
 - Text processing (tokenization, stop-word removal)
 - Sentiment classification models (like VADER or ML models)
- A sentiment score is assigned to each product based on overall customer feedback.

This step helps in understanding real user opinions about the product.

6. Product Scoring

A scoring mechanism is developed to rank products objectively.

- Each product is assigned a score based on multiple factors:
 - Price (lower price = higher score)
 - Rating (higher rating = higher score)
 - Discount (higher discount = higher score)
 - Sentiment score (positive reviews = higher score)
- A weighted formula is used to combine these factors into a single score.

This ensures fair comparison across different platforms.

7. Recommendation

Based on the calculated scores, the system identifies the best product.

- Products are ranked from highest to lowest score
- The top-ranked product is recommended to the user
- The system may also suggest alternative options

This step provides intelligent and data-driven recommendations.

8. Result Display

Finally, the results are presented to the user through a clear and interactive interface.

- Display includes:
 - Product comparison table
 - Best recommended product
 - Price comparison across platforms

- Ratings and sentiment summary
- Visualization tools (charts, graphs) may be used for better understanding

This ensures the user can easily make a decision.

LIMITATIONS:

- Some websites may block data scraping.
- Product details may change frequently.
- APIs may not be available for all platforms.
- Product names may vary across platforms, causing matching issues.
- Review data may sometimes be inaccurate or biased.
- Real-time performance depends on internet and data availability.

CONCLUSION:

The AI-Based Multi-Platform Product Analysis and Recommendation System provides an efficient solution to simplify online shopping. It reduces the effort of manually comparing products across different platforms by combining data analytics and artificial intelligence.

The system helps users make smart decisions by analysing multiple factors such as price, rating, discount, and customer sentiment. This project is practical, user-friendly, and highly useful in real-world applications, making it a strong and valuable final-year project.